

CHAPTER 13: PLANT GROWTH AND DEVELOPMENT

Complete Master Study Guide & Comprehensive Academic Notes

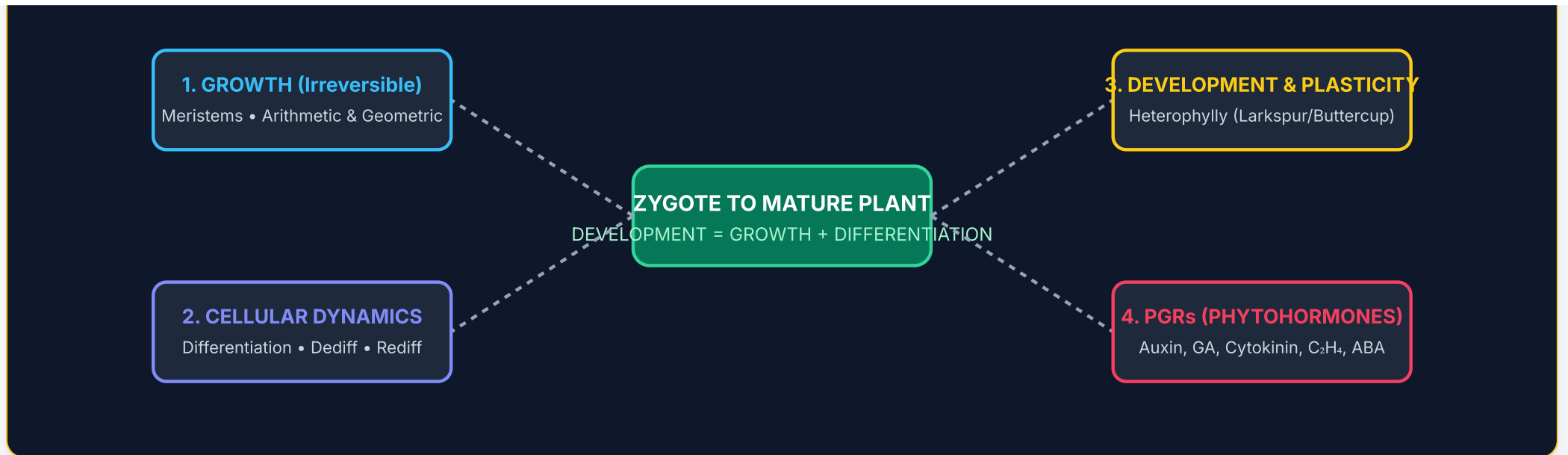
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13.1 GROWTH IN PLANTS

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📌 FUNDAMENTAL DEFINITION OF GROWTH

Growth is defined as an *irreversible permanent increase in size* of an organ or its parts or even of an individual cell. It is invariably accompanied by metabolic processes (both anabolic and catabolic) that occur at the expense of energy.

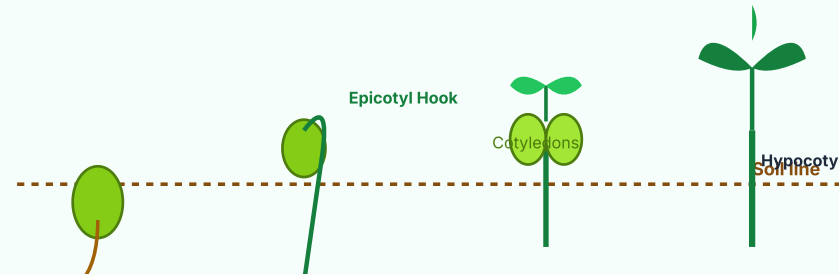
● First Step of Plant Growth: Seed Germination

The mature plant body develops from a single-celled **zygote (fertilised egg)** through a precise, highly ordered succession of events. The initial step is **seed germination**.

Favourable Conditions: When water, oxygen, and optimal temperature exist, the seed resumes metabolic activity and growth takes place.

Dormancy / Suspended Rest: In the absence of favourable conditions, seeds enter a period of suspended growth (rest/dormancy) until conditions normalize.

FIGURE 13.1 — GERMINATION AND SEEDLING DEVELOPMENT IN BEAN



● 13.1.1 Plant Growth Generally is Indeterminate

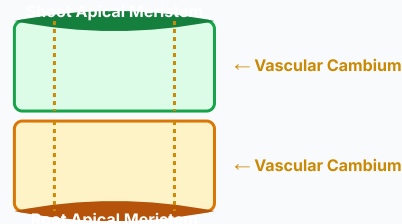
Plant growth is unique because plants retain the capacity for **unlimited/indeterminate growth** throughout their life. This capacity is due to the presence of **meristems** at specific locations in their body.

★ KEY CONCEPT: OPEN FORM OF GROWTH

The form of growth wherein *new cells are continuously added to the plant body by the activity of meristems* is termed the **open form of growth**.

Meristem Type	Location	Contribution & Role
Root Apical Meristem (RAM)	Root apex / tips	Primary Growth: Elongation of the plant axis along length.
Shoot Apical Meristem (SAM)	Shoot apex / tips	Primary Growth: Main shoot axis lengthening & organ formation.
Lateral Meristems (Vascular Cambium & Cork Cambium)	Lateral sides of stems & roots in Dicots/Gymnosperms	Secondary Growth: Increases the girth/diameter of plant organs later in life.

Figure 13.2 — Diagrammatic Representation of Meristem Locations



13.1.2 Growth is Measurable

At a cellular level, growth is principally a consequence of an **increase in the amount of protoplasm**. Direct protoplasm measurement is difficult, so growth is measured by proportional parameters:

1. Fresh Weight & Dry Weight

2. Length (e.g., Pollen Tube)

3. Surface Area (Dorsiventral Leaf)

4. Volume & Cell Number

NCERT QUANTITATIVE EXAM FACTS

- **Maize Root Apical Meristem:** Produces more than **17,500 new cells per hour** (expressed as increase in *cell number*).
- **Watermelon Cells:** May increase in size by up to **3,50,000 times** (expressed as increase in *cell size*).

13.1.3 Phases of Growth

The period of growth is divided into three distinct chronological and anatomical phases (exemplified by root tips):

Phase	Cellular Location	Structural Characteristics
1. Meristematic Phase	Root & shoot apices (constantly dividing tip)	Rich in protoplasm, large conspicuous nuclei, thin primary cellulosic walls, abundant plasmodesmata.

Phase	Cellular Location	Structural Characteristics
2. Elongation Phase	Proximal to meristematic zone	Increased vacuolation, cell enlargement, new cell wall deposition.
3. Maturation Phase	Proximal to elongation zone (further from apex)	Attain maximal size, cell wall thickening, complex protoplasmic modifications (differentiated tissues).

13.1.4 GROWTH RATES: ARITHMETIC VS GEOMETRIC

Mathematical Dynamics

Growth Rate is defined as the *increased growth per unit time*. Organisms or organs produce new cells via two distinct mathematical patterns:

A. Arithmetic Growth

Linear Curve

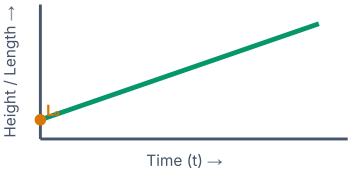
Following mitotic cell division, **only one daughter cell continues to divide**, while the other differentiates and matures. Exemplified by a root elongating at a constant rate.

Mathematical Formula:

$$L_t = L_0 + r t$$

- L_t = Length at time 't'
- L_0 = Length at time 'zero'
- r = Growth rate / elongation per unit time

Plot of Length (L) vs Time (t)



B. Geometric Growth

Sigmoid (S) Curve

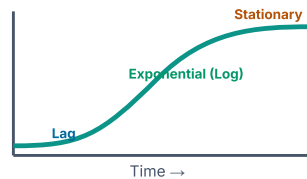
Following mitotic cell division, **both progeny cells retain the ability to divide**. Growth is initial slow (lag), then exponential (log), and finally slows to a stationary phase due to limited nutrient supply.

Mathematical Formula:

$$W_1 = W_0 e^{rt}$$

- W_1 = Final size (weight, height, number)
- W_0 = Initial size at beginning of period
- r = Growth rate / Relative growth rate (Efficiency Index)
- t = Time of growth | e = Base of natural logarithms

Sigmoid Growth Curve (S-Curve)



Comparative Analysis: Absolute vs Relative Growth Rate (Figure 13.7 Analysis)

Absolute Growth Rate

Measurement and comparison of *total growth per unit time*.

Relative Growth Rate

Growth of the given system per unit time expressed on a *common basis (per unit initial parameter)*.

NCERT Leaf Experiment Case:

- **Leaf A:** $5\text{ cm}^2 \rightarrow 10\text{ cm}^2$ (Abs growth = 5 cm^2 , Rel growth = **100%**)
 - **Leaf B:** $50\text{ cm}^2 \rightarrow 55\text{ cm}^2$ (Abs growth = 5 cm^2 , Rel growth = **10%**)
- Leaf A shows a much higher relative growth rate!*

1. Water

Required for cell enlargement (turgidity) & enzymatic medium.

2. Oxygen

Helps release metabolic energy essential for growth activities.

3. Nutrients

Macro & micro elements synthesized into protoplasm & energy.

4. Temperature

Optimum range best suited; deviations are detrimental.

5. Light & Gravity

Environmental signals affecting stages/directions of growth.

13.2

DIFFERENTIATION, DEDIFFERENTIATION AND REDIFFERENTIATION

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Cells derived from root apical meristem, shoot apical meristem, and cambium undergo structural and functional maturation. Differentiation in plants is also **open** because cells from the same meristem mature into different structures based on their location.

Phenomenon	Biological Definition	Structural / Protoplasmic Changes	Textbook Examples
DIFFERENTIATION	The act leading to maturation of meristematic cells to perform specific specialized functions.	Loss of protoplasm (in tracheary elements); development of strong, elastic, lignocellulosic secondary cell walls.	Formation of tracheary elements , vessel elements, root cap cells, epidermal cells.
DEDIFFERENTIATION	The phenomenon where living, fully differentiated cells that have lost division capacity <i>regain the capacity to divide</i> under specific conditions.	Re-activation of mitotic cell division capacity; reversal to meristematic state.	Formation of interfascicular cambium and cork cambium (phellogen) from fully differentiated parenchyma cells.
REDIFFERENTIATION	The act of cells produced by dedifferentiated meristems <i>losing division capacity once again</i> and maturing to perform specific functions.	Secondary cell wall thickening, specialized protoplasmic modifications for secondary tissues.	Formation of secondary xylem , secondary phloem , cork (phellem) , and phelloderm .

POSITIONAL CONTROL OF DIFFERENTIATION

The final structure at maturity of a cell/tissue is determined by its **location within the plant body**:

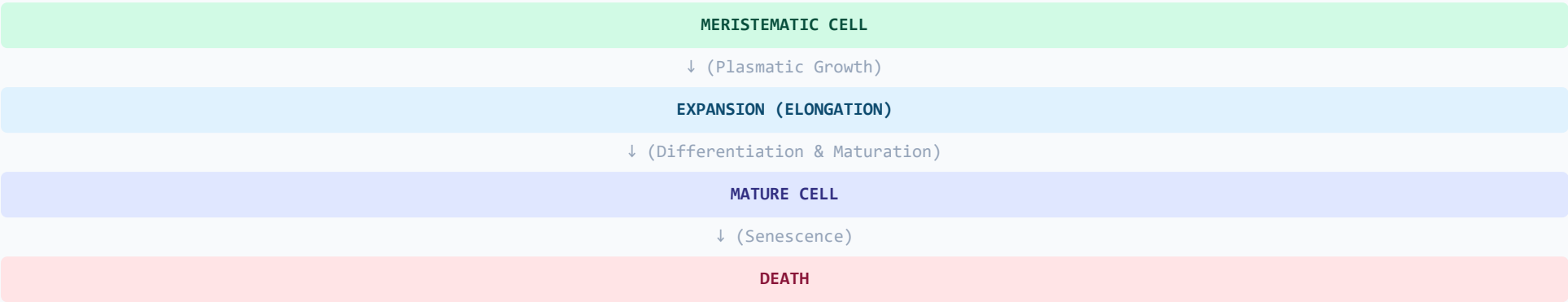
- Cells positioned *away from root apical meristem* → Differentiate into **Root-cap cells**.
- Cells pushed to the *periphery* → Mature into **Epidermal cells**.

DEFINITION OF DEVELOPMENT

Development includes all changes that an organism goes through during its life cycle from seed germination to senescence.

DEVELOPMENT = GROWTH + DIFFERENTIATION

Figure 13.8 — Sequence of Development in a Plant Cell



 PLASTICITY & HETEROPHYLLY

Plasticity: The ability of plants to follow different pathways in response to environment or phases of life to form different kinds of structures. The presence of different leaf shapes on the same plant is termed **Heterophyly**.

Type of Heterophyly	Triggering Factor	Textbook Plant Examples	Morphological Differences
Developmental Plasticity	Phase of Life (Juvenile vs Adult)	Cotton, Coriander, Larkspur	Leaves of juvenile plant are distinctly different in shape from those of mature/adult plant.
Environmental Plasticity	Environmental Habitat (Air vs Water)	Buttercup (Ranunculus)	Broad, lobed leaves produced in air; highly dissected, thread-like leaves produced in water.

 MEMORY TRICK: HETEROPHYLLY EXAMPLES

"Cotton, Coriander, Larkspur = Childhood to Life (Developmental!)"
"Buttercup floats in Water & Air (Environmental!)"

13.4 PLANT GROWTH REGULATORS (PGRs / PHYTOHORMONES)

NCERT Pages 174–178

13.4.1 CHEMICAL COMPOSITION & FUNCTIONAL GROUPING

PGR Group	Chemical Nature / Derivative	Specific Chemical Examples	Functional Classification
Auxins	Indole Compounds	Indole-3-acetic acid (IAA)	Plant Growth Promoter
Gibberellins	Terpenes	Gibberellic acid (GA ₃)	Plant Growth Promoter
Cytokinins	Adenine (Purine) Derivatives	N ⁶ -furfurylamino purine (Kinetin), Zeatin	Plant Growth Promoter
Abscisic Acid (ABA)	Carotenoid Derivatives	Abscisic Acid	Plant Growth Inhibitor (Stress)
Ethylene	Gaseous Hydrocarbon	Ethylene (C ₂ H ₄)	Largely Inhibitor (Dual Action)

13.4.2 HISTORICAL DISCOVERIES (ALL 5 ACCIDENTAL DISCOVERIES)

1. AUXIN

Charles & Francis Darwin: Canary grass coleoptiles bend toward unilateral light (Phototropism).

F.W. Went: Isolated Auxin from oat (*Avena*) seedling coleoptile tips.

2. GIBBERELLIN

'Bakanae' (Foolish Seedling): Rice disease caused by fungus *Gibberella fujikuroi*.

E. Kurosawa (1926): Treated rice with sterile fungal filtrate → Active substance GA₃.

3. CYTOKININ

F. Skoog & co-workers: Tobacco stem callus proliferated only with auxin + yeast extract/coconut milk/DNA.

Miller et al. (1955): Crystallized active compound → **Kinetin**.

4. ABSCISIC ACID

Mid-1960s: 3 independent groups discovered Inhibitor-B, Abscission II, and Dormin.

All proved chemically identical → Named **Abscisic Acid (ABA)**.

5. ETHYLENE

H.H. Cousins (1910): Volatile substance from ripe oranges hastened ripening of stored bananas.

Identified as gaseous hormone **Ethylene**.

13.4.3.1 AUXINS

Greek 'auxein' = to grow

Natural Auxins: Indole-3-acetic acid (IAA), Indole butyric acid (IBA).

Synthetic Auxins: Naphthalene acetic acid (NAA), 2,4-Dichlorophenoxyacetic acid (2,4-D).

First isolated from human urine. Produced at growing shoot & root apices.

KEY PHYSIOLOGICAL ROLES & APPLICATIONS:

- **Rooting in Stem Cuttings:** Widely used in horticultural plant propagation.
- **Apical Dominance:** Apical bud inhibits lateral (axillary) buds. Decapitation (removal of shoot tip) allows lateral growth (applied in *tea plantations* & *hedge-making*).
- **Fruit & Leaf Abscission:** Prevents premature drop of young leaves/fruits; promotes abscission of older mature leaves/fruits.
- **Parthenocarp:** Induces seedless fruit development in tomatoes.
- **Selective Herbicide:** 2,4-D kills broad-leaved dicot weeds without affecting mature monocot plants (used for weed-free lawns).
- **Vascular Control:** Controls xylem differentiation and aids cell division.

13.4.3.2 GIBBERELLINS (GAs)

> 100 Types (GA₁, GA₂, GA₃...)

All Gibberellins are **acidic**. **GA₃ (Gibberellic Acid)** was one of the first discovered and remains the most intensively studied.

KEY PHYSIOLOGICAL ROLES & APPLICATIONS:

- **Stem & Axis Elongation:** Used to increase the length of grape stalks.
- **Fruit Shaping & Size:** Causes fruits like apple to elongate and improve shape.
- **Delaying Senescence:** Fruits can be left on trees longer to extend market period.
- **Brewing Industry:** GA₃ speeds up the malting process in barley.
- **Sugarcane Yield Enhancement:** Increases stem internode length, increasing yield by up to **20 tonnes per acre**.
- **Juvenile Conifer Hastening:** Hastens maturity for early seed production.
- **Bolting Effect:** Promotes internode elongation just prior to flowering in beet, cabbages, and rosette plants.

13.4.3.3 CYTOKININS

Discovered as **Kinetin** from autoclaved herring sperm DNA (does not occur naturally). Natural cytokinin: **Zeatin** (isolated from corn-kernels & coconut milk).

KEY ROLES:

- Promotes **cytokinesis** (cell division) in root apices, shoot buds, young fruits.
- Forms new leaves & chloroplasts in leaves.
- Promotes adventitious & lateral shoot growth.
- **Overcomes Apical Dominance.**
- **Delays Leaf Senescence** by nutrient mobilization.

13.4.3.4 ETHYLENE (C₂H₄)

Simple **gaseous PGR**. Commercial source: **Ethephon** (aqueous solution releasing ethylene slowly).

KEY ROLES:

- **Fruit Ripening:** Causes *respiratory climacteric* (sharp rise in respiration rate).
- **Triple Response in Seedlings:** Horizontal growth, swelling of axis, apical hook in dicot seedlings.
- **Breaks Dormancy:** Seeds/buds, peanut germination, potato tuber sprouting.
- **Deep-Water Rice:** Rapid petiole/internode elongation keeping shoot above water.

- **Root Growth & Hair Formation:** Increases absorption surface area.
- **Ethephon Application:** Thinning of cotton/walnut, promotes female flowers in cucumbers.

13.4.3.5 **ABSCISIC ACID (ABA)**

Known as the **"STRESS HORMONE"**. General growth inhibitor and antagonist to Gibberellins (GAs).

KEY ROLES:

- **Inhibits Seed Germination** and plant metabolism.
- **Stomatal Closure:** Stimulates stomatal closure under water stress/drought.
- **Seed Dormancy & Maturation:** Helps seeds withstand desiccation and unfavourable growth factors.
- **Antagonist to GA:** GA promotes germination; ABA enforces dormancy.

 MASTER COMPARATIVE MATRIX: ALL 5 PLANT GROWTH REGULATORS

PGR Parameter	Auxins	Gibberellins (GA)	Cytokinins	Ethylene	Abscisic Acid (ABA)
Primary Function	Cell elongation, apical dominance	Stem elongation, bolting, malting	Cell division, delay senescence	Fruit ripening, abscission	Stress response, seed dormancy
Apical Dominance Effect	Promotes (Inhibits axillary buds)	No direct effect	Overcomes (Promotes lateral buds)	No direct effect	No direct effect
Stomatal Action	No direct control	No direct control	Opens stomata	No direct control	Closes Stomata (Stress)
Seed Germination	Minor role	Promotes (Breaks dormancy)	Promotes	Breaks peanut seed dormancy	Inhibits (Enforces dormancy)

PGR Parameter	Auxins	Gibberellins (GA)	Cytokinins	Ethylene	Absciscic Acid (ABA)
Agricultural Key Use	Rooting cuttings, 2,4-D herbicide	Sugarcane yield (+20 tons/acre)	Tissue culture callus growth	Ethephon (Fruit ripening)	Drought resistance study

13.6 NCERT TEXTBOOK EXERCISES & SOLUTIONS

Pages 179–180

Q1. Define growth, differentiation, development, dedifferentiation, redifferentiation, determinate growth, meristem and growth rate.

- **Growth:** Irreversible permanent increase in size of an organ or cell accompanied by metabolic activity.
- **Differentiation:** Process by which meristematic cells mature to perform specific functions by undergoing wall/protoplasmic changes.
- **Development:** Sum total of growth and differentiation covering the complete life cycle from seed germination to senescence.
- **Dedifferentiation:** Process where living mature differentiated cells regain the capacity to divide (e.g., formation of cork cambium).
- **Redifferentiation:** Act where cells produced by dedifferentiated meristems lose division capacity again and mature for specialized functions.
- **Determinate Growth:** Growth that stops after reaching a certain genetically fixed size and shape (e.g., leaves, flowers, fruits).
- **Meristem:** Specialized regions of active cell division in plants containing self-perpetuating undifferentiated cells.
- **Growth Rate:** Increased growth per unit time, expressed mathematically as arithmetic or geometric growth.

Q2. Why is not any one parameter good enough to demonstrate growth throughout the life of a flowering plant?

Answer: Growth at the cellular level is an increase in protoplasm, which cannot be directly measured. Different plant organs undergo growth in different dimensions—e.g., root/shoot apex growth is measured by cell division (number), watermelon growth by cell volume, pollen tube growth by length, and leaf growth by surface area. Furthermore, fresh weight varies with water content, while dry weight requires plant destruction. Hence, no single parameter suffices.

Q3. Describe briefly: (a) Arithmetic growth (b) Geometric growth (c) Sigmoid growth curve (d) Absolute and relative growth

Answer:

(a) **Arithmetic Growth:** Only one daughter cell divides post-mitosis; gives linear curve ($L_t = L_0 + rt$).

- (b) **Geometric Growth:** Both daughter cells divide post-mitosis; initial lag phase followed by exponential log phase ($W_1 = W_0 e^{rt}$).
- (c) **Sigmoid Growth Curve:** S-shaped curve of living organisms in nature comprising Lag, Log (Exponential), and Stationary phases.
- (d) **Absolute vs Relative Growth:** Absolute growth is total growth per unit time; Relative growth is growth rate per unit initial parameter.

Q4. List five main groups of natural plant growth regulators. Write a note on discovery, physiological functions and agricultural applications of any one of them.

Answer: Five groups: Auxins, Gibberellins, Cytokinins, Ethylene, Absciscic Acid.

Note on Auxin: Discovered accidentally by Charles & Francis Darwin (canary grass phototropism) and isolated by F.W. Went from oat coleoptiles. Functions: Apical dominance, rooting cuttings, cell elongation, parthenocarpy. Application: 2,4-D used as selective weedicide for dicot-free lawns.

Q5. Why is abscisic acid also known as stress hormone?

Answer: ABA is termed the stress hormone because it stimulates rapid **closure of stomata** during water stress (drought) to prevent transpiration loss and enhances plant tolerance to environmental stresses by inducing seed dormancy to survive desiccation.

Q6. 'Both growth and differentiation in higher plants are open'. Comment.

Answer: Growth is open because meristems continuously add new cells throughout the plant's life. Differentiation is open because cells arising from the exact same meristem differentiate into different structural tissues at maturity based on their spatial location (e.g., peripheral cells become epidermis while distal cells become root cap).

Q7. Which one of the plant growth regulators would you use if you are asked to:

(a) Induce rooting in a twig → **Auxin (IBA / NAA)**

(b) Quickly ripen a fruit → **Ethylene (Ethephon)**

(c) Delay leaf senescence → **Cytokinin**

(d) Induce axillary bud growth → **Cytokinin**

(e) 'Bolt' a rosette plant → **Gibberellin (GA_3)**

(f) Immediate stomatal closure → **Abscisic Acid (ABA)**

Q8–Q10. Expected physiological outcomes:

- **GA_3 applied to rice seedlings:** Causes rapid stem elongation ('Bakanae' foolish seedling hyper-elongation effect).

- **Dividing cells stop differentiating:** Plant tissue forms an undifferentiated mass of cells called a **callus** without developing functional organs.
- **Rotten fruit mixed with unripe fruits:** Volatile **ethylene** released from rotten fruit accelerates rapid ripening of unripe fruits.
- **Forgetting Cytokinin in culture medium:** Cell division and shoot bud differentiation from callus will fail, as cytokinin is required along with auxin for cell division.



FINAL QUICK REVISION SUMMARY

Rapid Exam Recap

Growth Equations

- Arithmetic: $L_t = L_0 + rt$ (Linear)
- Geometric: $W_1 = W_0 e^{rt}$ (Sigmoid/S-Curve)

Cell Transformations

- Meristem $\rightarrow$ Mature = Differentiation
- Mature $\rightarrow$ Meristem = Dedifferentiation
- Dediff Meristem $\rightarrow$ Mature = Redifferentiation

Plasticity / Heterophylly

- Developmental: Cotton, Coriander, Larkspur
- Environmental: Buttercup (Air vs Water)

Hormonal Antagonism

- ABA vs GA: Seed Dormancy vs Germination
- Auxin vs Cytokinin: Apical Dominance vs Lateral Growth